



Deliverable 3: Process Modeling for FindMyMeal

IS436 Structured Systems Analysis and Design
10/27/2024

Team: FindMyMeal

Team Members:

Hunnain Arsalan - Team Lead / Frontend Developer

Muhammad Wajahat - Project Manager

Benaya Berhane - Database Developer

Bazen Mekonen - Backend Developer

Ishan Giri - Networking Specialist

Michael Glass - Database Developer, Testing

Introduction

In Deliverable 3, we focus on Process Modeling to provide a detailed analysis of the data flows in our project, FindMyMeal.

Objective:

- Develop and present Data Flow Diagrams (DFDs) that illustrate the system's data interactions.
- Refine our requirements and use cases based on these models, and update the project plan to reflect recent progress.

Key Components:

- Context Diagram
- Level 0 and Level 1 DFDs
- Revised Requirements and Use Cases
- Updated Project Plan

This deliverable helps ensure our system design aligns with user needs and project objectives.

Data Flow Diagrams (DFDs) Overview

Data Flow Diagrams (DFDs) illustrate the flow of information within our system.

Purpose of DFDs:

- To visually model how data moves within and between processes, entities, and data stores.
- Ensures that all parts of the system interact logically and efficiently.

Levels of DFDs:

- Context Diagram: Shows the entire system and its external interactions.
- Level 0 DFD: Breaks down the system into main processes.
- Level 1 DFD: Further decomposes each Level 0 process for more detail.

These diagrams help us visualize and confirm that the system handles data flows effectively.

Context Diagram



The Context Diagram represents the FindMyMeal system as a single process and highlights interactions with external entities.



Entities



- User: Interacts with the system to create accounts, log in, search for recipes, and save recipes.



- Recipe Database: Stores recipes and handles user queries based on input criteria (ingredients or nutritional values).

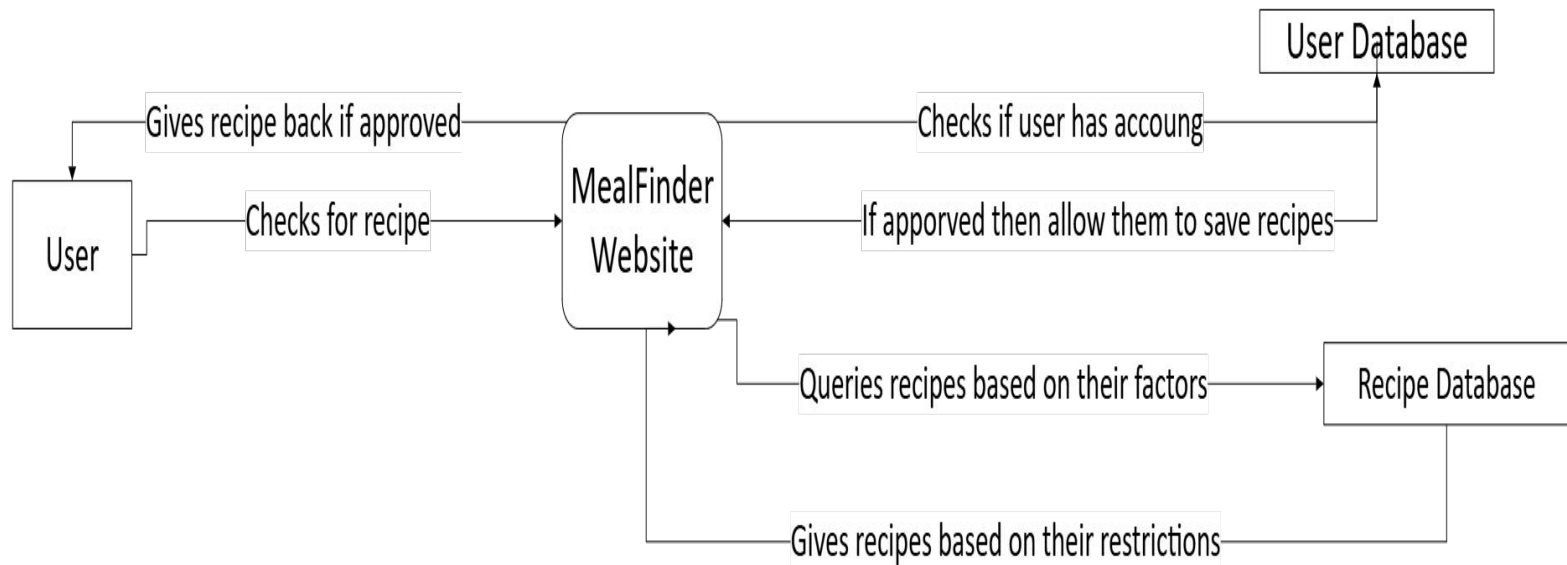


- User Database: Manages user data, including account information and saved recipes.



The Context Diagram helps outline high-level interactions without internal complexities.

Context Diagram





Level 0 DFD

The Level 0 DFD decomposes the system into four primary processes:

1. Account Creation

- Allows new users to create an account with their credentials.

2. Login

- Validates user credentials for access to account and saved recipes.

3. Recipe Search

- Allows users to search recipes by ingredients or nutrition; results are fetched from the Recipe Database.

4. Save Recipe

- Users save recipes to their account for future reference.

This diagram shows interactions with the Recipe and User databases and highlights main data flows.

An abstract background on the left side of the slide. It features a grid of glowing, semi-transparent squares in shades of blue, green, and orange. Overlaid on this grid are several white arrows of varying sizes and orientations, some pointing right and some pointing down. A prominent white arrow points down from the top left, and another points right from the middle left. A large, glowing white arrow points down from the middle left, and another points right from the middle right. The overall effect is a sense of digital flow and movement.

Level 1 DFDs: Account Creation and Login

The Level 1 DFDs for Account Creation and Login break down the internal steps in each process.

Account Creation:

- User provides email, password, and username.
- The system checks for an existing account; if none exists, the account is created and stored.

Login:

- User enters email and password.
- System verifies credentials in the User Database, granting access if valid.

These steps ensure secure and efficient account management.

Level 1 DFDs: Recipe Search and Save Recipe

The Level 1 DFDs for Recipe Search and Save Recipe illustrate detailed steps for these core functionalities.

Recipe Search:

- User inputs criteria (ingredients or nutrition), and the system queries the Recipe Database.
- Results are displayed based on matching recipes or close alternatives if no exact match.

Save Recipe:

- User selects a recipe to save, which is then stored in their account within the User Database.

These processes ensure users can find and retain recipes based on personal preferences.



Descriptions of DFD Components

Detailed descriptions of each component in the DFDs:

Processes:

- Account Creation: New user account creation process.
- Login: Verification of user credentials.
- Recipe Search: Search functionality by ingredients or nutrition.
- Save Recipe: Storing selected recipes for future access.

Updated Requirements Definition

Based on DFD insights, we updated our requirements.

Functional Requirements:

- Account Management: User can create accounts and log in.
- Recipe Search: Users search by ingredients or nutrition.
- Recipe Saving: Users save recipes for future reference.

Non-Functional Requirements:

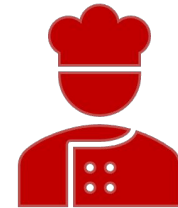
- Performance: Recipes displayed within 3 seconds.
- Security: Data encrypted; access restricted to

Revised Use Cases

Five main use cases based on system functionality:

1. Account Creation
2. Login
3. Recipe Search by Ingredients
4. Recipe Search by Nutrition
5. Save Recipe

Each use case includes preconditions, triggers, normal and alternative courses, and post-conditions to outline specific actions.



Project Plan Update

The project plan is updated to reflect progress:

- Requirements Gathering: Completed
- System Design (DFDs): 10/15/2024 - 10/28/2024
- Development Phase: 10/29/2024 - 11/25/2024
- Testing Phase: 11/26/2024 - 12/10/2024
- Deployment: 12/11/2024 - 12/15/2024
- Final Presentation & Report: 12/16/2024 - 12/20/2024

Team Information

- Team Name: FindMyMeal
- Roles and Contact:
 - Hunnain Arsalan: Team Lead / Frontend Developer
 - Muhammad Wajahat: Project Manager
 - Benaya Berhane: Database Developer
 - Bazen Mekonen: Backend Developer
 - Ishan Giri: Networking Specialist
 - Michael Glass: Database Developer, Testing
- Meeting Schedule: Mondays and Wednesdays at 4 pm.

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- **Deliverable 3** showcases detailed DFD modeling, updated requirements, revised use cases, and an updated project plan.

- Focus on development, followed by testing and deployment phases.

- We welcome any questions or feedback on the deliverable and project plan.